TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Boiling Point of a Liquid by Platinum Resistance Thermometer

Platinum's resistance climbs smoothly and repeatably with temperature. Calibrate the coil in ice and steam, dip it in the boiling liquid, and its resistance states the boiling point.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 17 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the boiling point of a liquid (aniline) with a platinum resistance thermometer (PRT). The resistance of a pure platinum coil varies with temperature t as

$$R_t = R_0\,(1 + \alpha\, t)\,, \quad \alpha \approx 3.9 \times 10^{-3}\ ^\circ\mathrm{C}^{-1}$$

where R_0 is the resistance in melting ice. Measuring the coil in ice (R_0), in steam (R_{100}) and in the boiling liquid (R_t) eliminates α explicitly — the **platinum temperature** is defined by the linear interpolation

$$t = (R_t - R_0) / (R_{100} - R_0) \times 100$$

The resistance is measured with a Wheatstone bridge, the coil forming one arm; at balance $P/Q = R_t/R_{box}$, so R_t follows from the known ratio and the resistance-box setting. (Strictly, R_t follows a shallow quadratic, so the platinum temperature differs from the gas-scale temperature by a small correction that grows beyond a few hundred degrees; near 180 °C it is under a degree and is neglected here, but it must be mentioned in an honest report.)

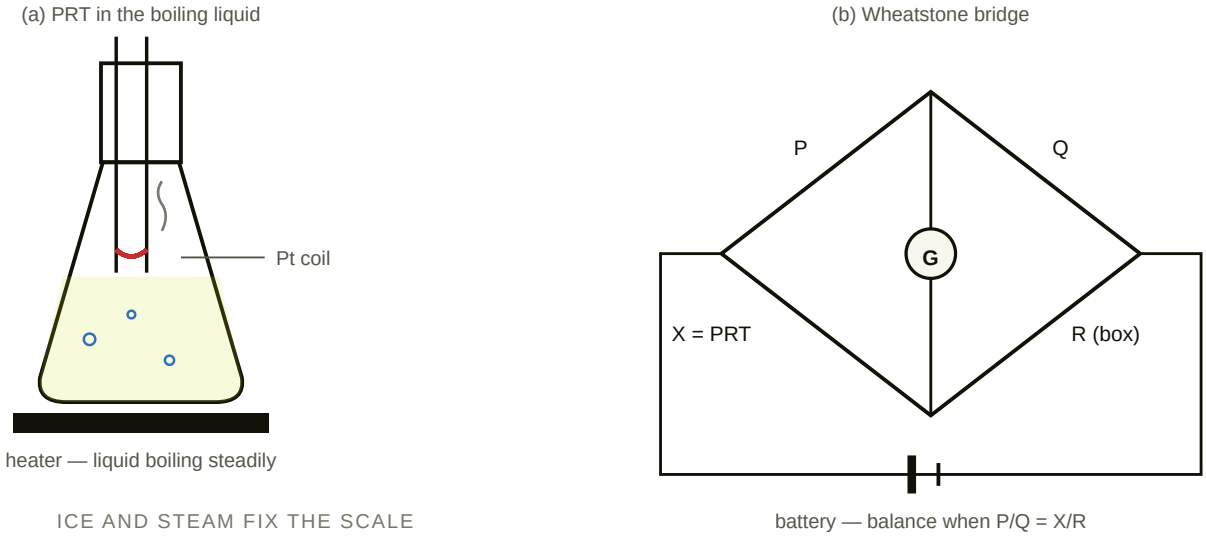


Figure 1 — (a) The platinum coil, sealed in its sheath, immersed in the steadily boiling liquid; its leads run to (b) the Wheatstone bridge, where the coil's resistance X is measured against a resistance box R with ratio arms P and Q.

2. Apparatus and procedure

- **Apparatus:** platinum resistance thermometer, Wheatstone bridge (or metre bridge) with resistance box, galvanometer, battery and keys, ice bath, steam hypsometer, flask of aniline with heater, barometer.
- Connect the coil as arm X of the bridge with equal ratio arms (P = Q), so R_t equals the box reading at balance. Use compensating leads, or keep the lead arrangement identical for all three measurements, so lead resistance cancels in the differences.
- Ice point: pack the coil's sheath in melting ice, wait for the reading to settle, balance the bridge and record R_0.
- Steam point: place the coil in steam above boiling water in the hypsometer (not in the water), record the barometric pressure for the true steam temperature, balance and record R_{100}.
- Boiling liquid: immerse the coil in aniline boiling steadily — the bulb fully covered, the boil neither violent nor dying — balance and record R_t. Repeat after a few minutes to confirm the value is steady.
- Compute t from the interpolation formula; compare with the accepted boiling point of aniline, 184 °C at standard pressure, noting the day's barometric pressure if a refined comparison is wanted.

3. Worked calculation

MEASUREMENT	RESISTANCE
Coil in melting ice, R_0	10.00 Ω

Coil in steam, R_{100}	13.90 Ω
Coil in boiling aniline, R_t	17.18 Ω

$$\alpha = (13.90 - 10.00) / (10.00 \times 100) = 3.90 \times 10^{-3}\ ^\circ C^{-1}$$

$$t = (17.18 - 10.00) / (13.90 - 10.00) \times 100 = 7.18 / 3.90 \times 100 \approx 184\ ^\circ C$$

RESULT

Boiling point of aniline ≈ 184 °C — agreeing with the accepted value of 184 °C at standard atmospheric pressure. The measured α, $3.90 \times 10^{-3}\ ^\circ C^{-1}$, also matches the textbook value for pure platinum, a good internal check that the coil and the bridge work are sound.

4. Precautions

- The bridge current heats the coil slightly (self-heating): use the smallest current that gives a readable deflection, close the key only briefly, and check that the balance does not drift while the key is held.
- The coil must be fully immersed in the boiling liquid and the liquid must be at a steady rolling boil; a half-boiled or superheated liquid reads falsely.
- In steam, the coil must be bathed in vapour, not splashed with water, and the steam temperature follows the day's pressure, not exactly 100 °C.
- Keep all connections tight and identical between the three measurements; a changing lead or contact resistance shifts every reading and does not cancel.

Practice (viva preparation)

1. Why is the thermometer calibrated at two fixed points instead of trusting the maker's value of α?
2. A coil reads R_0 = 25.00 Ω and R_{100} = 34.75 Ω. In an unknown bath it reads 44.50 Ω. Find the bath temperature (linear formula).
3. Why is platinum, rather than copper, the standard metal for resistance thermometry?
4. The bridge key is held down for a long time and the balance point creeps. Explain, and state the remedy.
5. What is meant by the "platinum temperature", and why does it differ slightly from the true (gas-scale) temperature at high t?

Answer key

1. α varies slightly from sample to sample with the purity and strain of the wire. Measuring R_0 and R_{100} on the actual coil fixes its own scale; the unknown temperature then comes from ratios in which any constant lead resistance and the exact α both cancel.
2. t = (44.50 − 25.00)/(34.75 − 25.00) × 100 = 19.50/9.75 × 100 = 200 °C.
3. Platinum is chemically inert, does not oxidise in air at high temperature, has a high melting point (1769 °C), and its resistance–temperature relation is exceptionally stable and reproducible; copper oxidises and its α varies more with sample history.
4. The measuring current heats the coil above the bath (self-heating), so its resistance creeps upward. Remedy: use a small current, tap the key only for an instant at balance, and confirm the reading is the same on a second quick balance.
5. The platinum temperature is the value given by the linear interpolation between the ice and steam resistances. The true R–t relation is slightly quadratic (a small negative second term), so above 100 °C the linear scale progressively misreads — by a fraction of a degree at 200 °C and several degrees at 500 °C — unless the quadratic (Callendar) correction is applied.

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